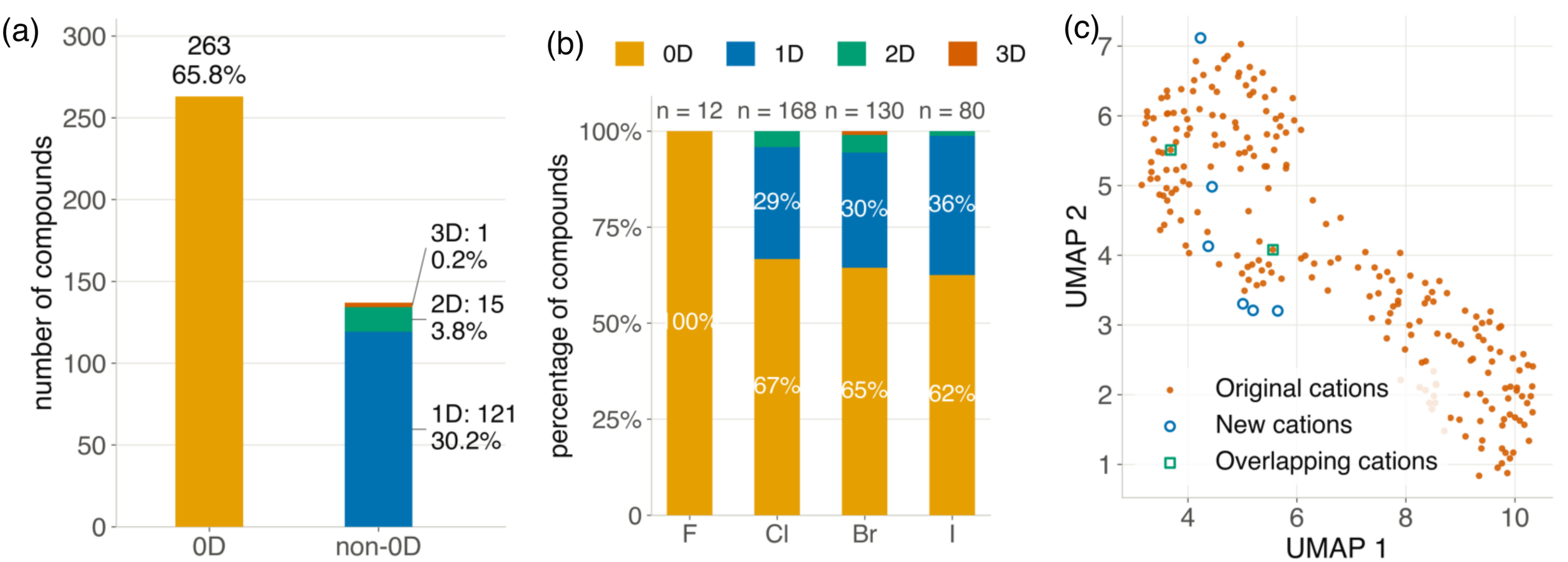

(a)
number of compounds
300
250
200
150
100
50
0
263
65.8%
3D: 1
0.2%
2D: 15
3.8%
1D: 121
30.2%
0D
non-0D
(b)
0D
1D
2D
3D
percentage of compounds
n = 12
n = 168
n = 130
n = 80
100%
75%
50%
25%
0
100%
29%
30%
36%
67%
65%
62%
F
Cl
Br
I
(c)
UMAP 2
UMAP 1
Original cations
New cations
Overlapping cations